

AI Multimedia Research Assistant System

1. Title of the Project:

AI Multimedia Research Assistant

2. Group Members:

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3. Introduction:

The **AI Multimedia Research Assistant** is proposed as an intelligent system that uses Artificial Intelligence to analyze multimedia content and convert it into structured and searchable knowledge. The system will allow users to upload multimedia content and automatically extract useful information such as transcripts, summaries, important topics, key points, notes, questions and answers, and relevant timestamps.

The system will also provide an **Ask AI** feature through which users can ask questions about their uploaded content and receive answers based on the information present in the multimedia files. This can be useful for students, researchers, professionals, teachers, and general users who need to quickly understand and retrieve information from large multimedia resources.

4. Objectives:

- 1.To develop an AI-based system that can analyze and understand multimedia content such as videos and audio recordings.
- 2.To automatically convert speech from multimedia files into text using Speech-to-Text technology.
- 3.To generate useful information such as summaries, key points, topics, notes, questions, and answers from the processed content.
- 4.To provide an intelligent search and Ask AI feature that allows users to retrieve specific information from their uploaded multimedia content.

5. Problem Statement

Large multimedia files often contain valuable information, but finding specific information from them is difficult and time-consuming. Users have to manually watch or listen to long videos or recordings to identify important topics or specific information.

Existing basic video or audio players mainly provide playback functionality and do not provide intelligent understanding, structured knowledge extraction, or context-based question answering. Therefore, there is a need for an AI-powered multimedia assistant that can process

long multimedia content, extract meaningful information, organize it into searchable knowledge, and allow users to interact with the content using natural language.

6. Proposed System / Methodology

The proposed system will provide a web-based platform where users can upload multimedia files such as videos and audio recordings. The system will process the uploaded content and generate structured information that can be searched and queried by the user.

Basic workflow: Multimedia Upload → Processing → Audio Extraction → Speech-to-Text → Transcript → AI Analysis → Knowledge Extraction → Search / Ask AI

Proposed Technology Stack

Layer	Technology
Frontend	React.js, TailwindCSS, Javascript,
Backend	Node.js, Express.js
AI Processing	Large Language Model (LLM), Embeddings, RetrievalAugmented Generation (RAG)
Database	MongoDB
Vector Database	ChromaDB & FAISS
Video Processing	FFmpeg
Authentication	JWT-based authentication and password hashing , bcrypt
Development Tools	VS Code, Git, GitHub, Postman, MongosCompass & Clusteler

7. Modules / Features

1. **User Authentication Module:** Provides registration and login functionality using secure authentication.
2. **Multimedia Upload Module:** Allows users to upload video and audio files for processing.
3. **Video Processing Module:** Extracts audio from uploaded video files and prepares them for further processing.
4. **Speech-to-Text Module:** Converts spoken content from audio/video into text using Speech-to-Text technology.

5. **AI Content Analysis Module:** Generates summaries, topics, key points, notes, and other useful information.
6. **AI Question & Answer Module:** Allows users to ask natural-language questions about their uploaded multimedia content.
7. **Smart Search Module:** Searches for topics or information across processed multimedia content.
8. **Knowledge Base / RAG Module:** Stores processed content in a searchable knowledge base and retrieves relevant information for user queries.
9. **Multimedia History & Dashboard:** Displays uploaded files, processing status, summaries, previous searches, and generated results.

8. Expected Outcome

- Reduce the time required to understand long videos and audio recordings.
- Automatically generate summaries and important information.
- Provide searchable transcripts and topics.
- Allow users to ask questions directly about uploaded content.
- Provide relevant timestamps for specific information.
- Help students and researchers create useful notes from educational content.

9. Conclusion

The **AI Multimedia Research Assistant** aims to solve the problem of extracting and retrieving useful information from large multimedia files. By combining Speech-to-Text, Artificial Intelligence, Natural Language Processing, semantic search, and RAG-based question answering, the system will transform unstructured multimedia content into useful and searchable knowledge.

10. References

1. OpenAI Whisper – Speech Recognition and Transcription Technology.
2. MongoDB Documentation – Database and Vector Search.
3. Node.js Documentation – Backend Development.
4. React.js Documentation – Frontend Development.
5. FFmpeg Documentation – Multimedia Processing.
6. Research papers and technical documentation related to Natural Language Processing, Large Language Models, Semantic Search, and Retrieval-Augmented Generation (RAG).